

Лабораторная работа №7

Дисциплина: Сетевые технологии

Пакавира Арсениу Висенте Луиш

2026-03-20

1 Цель работы

- Настройка службы DHCP
- Распределение адресов IPv4 и IPv6
- Анализ DHCP-трафика

2 Топология и захват трафика (IPv4 DHCP)

- Переименование устройств по шаблону
- PC1-bahis, bahis-sw-01, bahis-gw-01
- Включён захват трафика между коммутатором и маршрутизатором



Рисунок 1: Топология сети в GNS3 с переименованными устройствами и включённым захватом трафика

3 Системные параметры VyOS и пользователи

- Задано имя маршрутизатора
- Настроено доменное имя
- Создан пользователь bahis
- Удалён пользователь vyos

```
bahi@bahi-gw-01:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not pers
[edit]
bahi@bahi-gw-01# delete system login user vyos
[edit]
bahi@bahi-gw-01# commit
[edit]
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01#
```

4 Настройка DHCP-сервера (IPv4)

- Создан shared-network bahis
- DNS: 10.0.0.1
- Gateway: 10.0.0.1
- Диапазон: 10.0.0.2–10.0.0.253

```
1.netah1-gw-01# set service dhcp-server shared-network-name bahi domain-name ba
edit]
0.0.1ahi-gw-01# set service dhcp-server shared-network-name bahi name-server 10
edit]
/24@bahi-gw-01# set service dhcp-server shared-network-name bahi subnet 10.0.0.
edit]
/24 default-router 10.0.0.1 dhcp-server shared-network-name bahi subnet 10.0.0.
edit]
/24 range hosts start 10.0.0.2cp-server shared-network-name bahi subnet 10.0.0.
edit]
/24 range hosts stop 10.0.0.253p-server shared-network-name bahi subnet 10.0.0.
edit]
```

Рисунок 2: Конфигурация DHCP-сервера на маршрутизаторе VyOS для сети 10.0.0.0/24 6/22

5 Статистика DHCP-сервера (до клиентов)

- Всего адресов: 252
- Активных аренд: 0
- Все адреса доступны

```
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool      Size    Leases   Available  Usage
-----
bahi      252      0        252      0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  H
name
-----
bahi@bahi-gw-01:~$
```

Рисунок 3: Статистика DHCP-сервера и отсутствие активных аренд IP-адресов

6 DHCP-процесс на узле PC1 (Request / ACK)

- DHCP Request
- DHCP ACK
- Назначен IP 10.0.0.2/24

```
Option 12: Host Name = PC1-bah1
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = bah1.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = PC1-bah1

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
```

7 Проверка IP и связности PC1

- IP получен по DHCP
- Gateway: 10.0.0.1
- Ping маршрутизатора успешен

```
PC1-bahi> show ip
```

```
NAME       : PC1-bahi[1]  
IP/MASK    : 10.0.0.2/24  
GATEWAY    : 10.0.0.1  
DNS        : 10.0.0.1  
DHCP SERVER : 10.0.0.1  
DHCP LEASE  : 86359, 86400/43200/75600  
DOMAIN NAME : bahi.net  
MAC        : 00:50:79:66:68:00  
LPORT      : 20004  
RHOST:PORT  : 127.0.0.1:20005  
MTU        : 1500
```

8 DHCP-статистика после выдачи адреса

- Активных аренд: 1
- IP: 10.0.0.2
- Клиент: PC1-bahis

```
-vbash: /config/dhcpd.leases: Permission denied
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool      Size      Leases    Available  Usage
-----
bahi      252       0         252        0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Host
name
-----
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool      Size      Leases    Available  Usage
-----
bahi      252       1         251        0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Host
name
-----
```

9 Журнал DHCP-сервера

- DHCPDISCOVER
- DHCPOFFER
- DHCPREQUEST
- DHCPACK
- Обслуживание клиента PC1

```
bahi@bahi-gw-01:~$ show log | grep dhcp
Jan 23 13:44:50 sudo[2316]:      root : TTY=unknown ; PWD=/ ; USER=root ; COMMAND=/usr/bin/sh -c /usr/sbin/vysshim /usr/libexec/vyos/conf_mode/dhcp_server.py
Jan 23 13:44:50 vyos-configd[682]: Received message: {"type": "node", "data": "/usr/libexec/vyos/conf_mode/dhcp_server.py"}
Jan 23 13:44:58 dhcpd[2346]: Wrote 0 leases to leases file.
Jan 23 13:44:58 dhcpd[2346]: Lease file test successful, removing temp lease file: /config/dhcpd.leases.1769175898
Jan 23 13:44:58 dhcpd[2348]: Wrote 0 leases to leases file.
Jan 23 13:44:59 dhcpd[2348]:
Jan 23 13:44:59 dhcpd[2348]: No subnet declaration for eth2 (no IPv4 addresses).
Jan 23 13:44:59 dhcpd[2348]: ** Ignoring requests on eth2.  If this is not what
Jan 23 13:44:59 dhcpd[2348]: you want, please write a subnet declaration
Jan 23 13:44:59 dhcpd[2348]: in your dhcpd.conf file for the network segment
Jan 23 13:44:59 dhcpd[2348]: to which interface eth2 is attached. **
Jan 23 13:44:59 dhcpd[2348]:
Jan 23 13:44:59 dhcpd[2348]: No subnet declaration for eth1 (no IPv4 addresses).
Jan 23 13:44:59 dhcpd[2348]: ** Ignoring requests on eth1.  If this is not what
Jan 23 13:44:59 dhcpd[2348]: you want, please write a subnet declaration
Jan 23 13:44:59 dhcpd[2348]: in your dhcpd.conf file for the network segment
Jan 23 13:44:59 dhcpd[2348]: to which interface eth1 is attached. **
```

10 Анализ DHCP-трафика (IPv4)

- Discover → Offer → Request → ACK
- ARP и Gratuitous ARP
- ICMP Echo

Capturing from Standard input (bahi-Sw-01 Ethernet1 to bahi-gw-01 eth0)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
25	4910.768760	10.0.0.1	10.0.0.2	DHCP	342	DHCP ACK - Transaction ID 0x7de8f84e
26	4911.721949	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
27	4912.722803	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
28	4913.724446	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)
29	4979.699860	Private_66:68:00	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.2
30	4979.701380	0c:74:67:05:00:00	Private_66:68:00	ARP	60	10.0.0.1 is at 0c:74:67:05:00:00
31	4979.702425	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0x6f7c, seq=1/256, ttl=64 (reply in 32)
32	4979.704042	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0x6f7c, seq=1/256, ttl=64 (request in 31)
33	4980.704821	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) request id=0x707c, seq=2/512, ttl=64 (reply in 34)
34	4980.707176	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) reply id=0x707c, seq=2/512, ttl=64 (request in 33)
35	4984.863913	0c:74:67:05:00:00	Private_66:68:00	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
36	4984.864712	Private_66:68:00	0c:74:67:05:00:00	ARP	60	10.0.0.2 is at 00:50:79:66:68:00

> Frame 1: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits)

> Ethernet II, Src: 0c:74:67:05:00:00 (0c:74:67:05:00:00), Dst: IPv6mcast_16 (33:00:00:00:00:00)

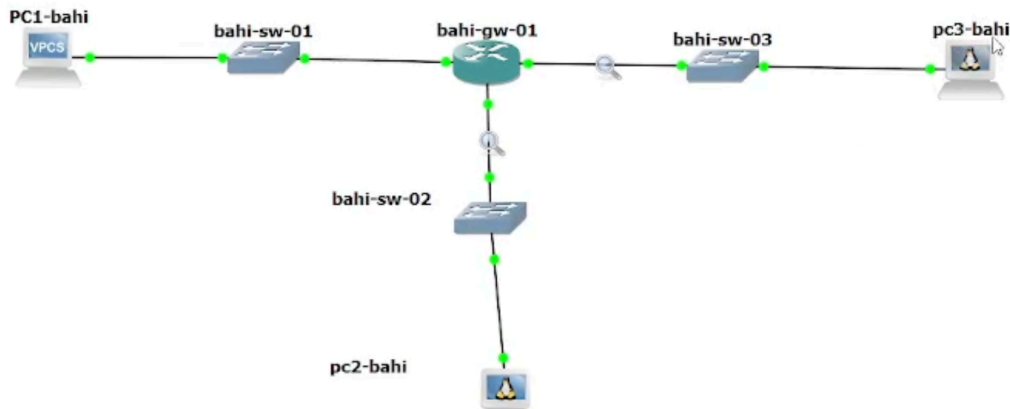
> Internet Protocol Version 6, Src: ::, Dst: ff02::16

> Internet Control Message Protocol v6

0000 33 33 00 00 00 16 0c 74 67 05 00 00 86 dd 60 00 33.....t g.....
0010 00 00 00 4c 00 01 00 00 00 00 00 00 00 00 00 00 ...L.....
0020 00 00 00 00 00 00 ff 02 00 00 00 00 00 00 00 00:.....
0030 00 00 00 00 00 16 3a 00 05 02 00 00 01 00 8f 00:.....
0040 69 4e 00 00 00 03 04 00 00 00 ff 02 00 00 00 00 iN.....
0050 00 00 00 00 00 01 ff 05 00 00 04 00 00 00 ff 05:.....
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

11 IPv6: топология и захват трафика

- Использован Alpine Linux
- Захват ICMPv6 и DHCPv6
- Подсети IPv6



12 Системные параметры VyOS (IPv6)

- Имя устройства
- Доменное имя
- Пользователь bahis

```
[edit]
vyos@vyos#
[edit]
vyos@vyos#
[edit]
vyos@vyos# set system domain-name bahi.net
[edit]
vyos@vyos#
[edit]
vyos@vyos# set system login user bahi authentication plaintext
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - bahi-gw-01 ttyS0

bahi-gw-01 login: bahi
Password:
```

13 Назначение IPv6-адресов интерфейсам

- eth1: 2000::1/64
- eth2: 2001::1/64

```
bahi@bahi-gw-01:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not persist until installed
[edit]
bahi@bahi-gw-01# delete system login user vyos
[edit]
bahi@bahi-gw-01# commit
[edit]
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
bahi@bahi-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
bahi@bahi-gw-01# show interfaces
  ethernet eth0 {
    hw-id 0c:fd:55:1c:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:fd:55:1c:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:fd:55:1c:00:02
  }
```

14 Stateless DHCPv6 + Router Advertisement

- SLAAC
- other-config-flag
- DNS и домен по DHCPv6

```
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01# set service router-advert interface eth1 prefix 2000::/64
[edit]
bahi@bahi-gw-01# set service router-advert interface eth1 other-config-flag
[edit]
bahi@bahi-gw-01# set service dhcpv6-server shared-network-name bahi-stateless
[edit]
bnet 2000::0/64# set service dhcpv6-server shared-network-name bahi-stateless su
[edit]
common-options name-server 2000::1pv6-server shared-network-name bahi-stateless co
[edit]
bahi@bahi-gw-01#
[edit]
common-options domain-search bahi.net-server shared-network-name bahi-stateless co
[edit]
```

15 Проверка IPv6 на PC2 (Stateless)

- SLAAC-адрес
- Link-local адрес
- Ping успешен

```
pc2-bah1 console is now available... Press RETURN to get started.  
/ # ip address  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000  
    link/ether 02:42:e8:69:53:00 brd ff:ff:ff:ff:ff:ff  
    inet6 fe80::42:e8ff:fe69:5300/64 scope link  
        valid_lft forever preferred_lft forever  
/ # ip -6 route show  
fe80::/64 dev eth0 metric 256  
/ # /# ping 2000::1 -c 2  
/bin/sh: /#: not found  
/ # ping 2000::1 -c 2  
PING 2000::1 (2000::1): 56 data bytes  
ping: sendto: Network unreachable  
/ # cat /etc/resolv.conf  
/ # udhcpd6 -i eth0  
udhcpd6: started, v1.37.0  
udhcpd6: sending discover  
udhcpd6: sending discover  
udhcpd6: sending discover  
udhcpd6: sending discover
```

16 Анализ IPv6-трафика (Stateless)

- Router Solicitation
- Router Advertisement
- DHCPv6 Solicit / Reply

capture.pcapng [alkamal-gw-01 eth1 to alkamal-sw-02 Ethernet1]

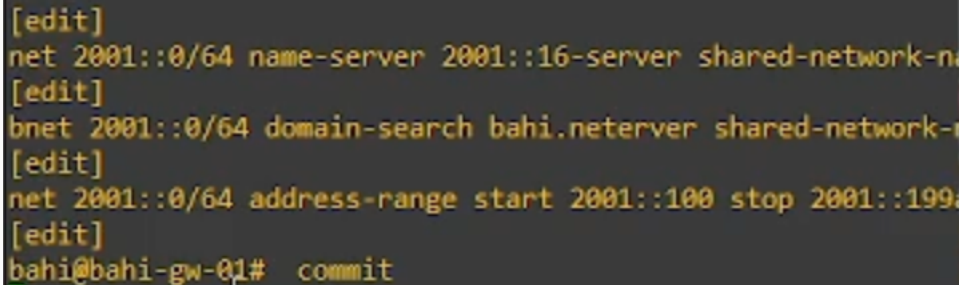
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
27	11..	2000::42:9bff:febb:db00	2000::1	ICMPv6	118	Echo (ping) request id=0x013c, seq=1, hop limit=64 (reply in 28)
28	11..	2000::1	2000::42:9bff:febb:db00	ICMPv6	118	Echo (ping) reply id=0x013c, seq=1, hop limit=64 (request in 27)
29	11..	fe80::edf:93ff:fe51	2000::42:9bff:febb:db00	ICMPv6	86	Neighbor Solicitation for 2000::42:9bff:febb:db00 from 0c:df:93:f5:00:01
30	11..	2000::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	78	Neighbor Advertisement 2000::42:9bff:febb:db00 (sol)
31	11..	fe80::42:9bff:fe51	fe80::edf:93ff:fe51	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fe51 from 02:42:9b:bb:db:00
32	11..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	ICMPv6	78	Neighbor Advertisement fe80::edf:93ff:fe51 (rtr, sol)
33	12..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	ICMPv6	86	Neighbor Solicitation for fe80::42:9bff:febb:db00 from 0c:df:93:f5:00:01
34	12..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	78	Neighbor Advertisement fe80::42:9bff:febb:db00 (sol)
35	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	116	Solicit XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
36	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Advertise XID: 0x81ba5f CID: 0003000102429bbdb00
37	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
38	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	134	Request XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
39	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x81ba6f CID: 0003000102429bbdb00
40	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
41	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	182	Request XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
42	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x81ba6f CID: 0003000102429bbdb00
43	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
44	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fe51 from 02:42:9b:bb:db:00
45	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	ICMPv6	78	Neighbor Advertisement fe80::edf:93ff:fe51 (rtr, sol)
46	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	ICMPv6	86	Neighbor Solicitation for fe80::42:9bff:febb:db00 from 0c:df:93:f5:00:01
47	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	78	Neighbor Advertisement fe80::42:9bff:febb:db00 (sol)
48	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	182	Request XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
49	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x81ba6f CID: 0003000102429bbdb00
50	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
51	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	182	Request XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
52	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x81ba6f CID: 0003000102429bbdb00
53	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
54	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	182	Request XID: 0x81ba6f [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
55	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x81ba6f CID: 0003000102429bbdb00
56	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
57	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	116	Solicit XID: 0x3e266e [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
58	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Advertise XID: 0x3e266e CID: 0003000102429bbdb00
59	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)
60	14..	fe80::42:9bff:febb:db00	ff02::1:2	DHCPv6	134	Request XID: 0x3e266e [ROOT-ONLY DOMAIN NAME]CID: 0003000102429bbdb00
61	14..	fe80::edf:93ff:fe51	fe80::42:9bff:febb:db00	DHCPv6	162	Reply XID: 0x3e266e CID: 0003000102429bbdb00
62	14..	fe80::42:9bff:febb:db00	fe80::edf:93ff:fe51	ICMPv6	210	Destination Unreachable (Port unreachable)

17 Настройка Stateful DHCPv6

- managed-flag
- Диапазон: 2001::100–2001::199
- DNS и домен

A terminal window with a dark background and yellow text. The text shows a series of configuration commands for Stateful DHCPv6. The commands are: [edit], net 2001::0/64 name-server 2001::16-server shared-network-n, [edit], bnet 2001::0/64 domain-search bahi.neterver shared-network-n, [edit], net 2001::0/64 address-range start 2001::100 stop 2001::199, [edit], and finally bahi@bahi-gw-01# commit. A green cursor bar is visible at the bottom left of the terminal window.

```
[edit]
net 2001::0/64 name-server 2001::16-server shared-network-n
[edit]
bnet 2001::0/64 domain-search bahi.neterver shared-network-n
[edit]
net 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
bahi@bahi-gw-01# commit
```

Рисунок 12: Конфигурация Stateful DHCPv6

18 Получение IPv6-адреса PC3

- До запроса: только link-local
- После udhcpc6: глобальный IPv6

```
PC3-alkamal console is now available... Press RETURN to get started.  
/ # ip address  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
10: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000  
    link/ether 02:42:88:32:29:00 brd ff:ff:ff:ff:ff:ff  
    inet6 fe80::42:88ff:fe32:2900/64 scope link  
        valid_lft forever preferred_lft forever  
/ # ip -6 route show  
fe80::/64 dev eth0 metric 256  
/ # cat /etc/resolv.conf  
/ # udhcpc6 -i eth0  
udhcpc6: started, v1.37.0  
udhcpc6: sending discover  
udhcpc6: sending select  
udhcpc6: IPv6 obtained, lease time 43200
```

Рисунок 13: Получение IPv6 по DHCPv6

19 Проверка IPv6 PC3 после DHCPv6

- Глобальный IPv6
- Default route
- DNS получен

```
/ # ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
10: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:88:32:29:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:88ff:fe32:2900/64 scope link
        valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0 metric 256
/ # ping 2001::1 -c 2
PING 2001::1 (2001::1): 56 data bytes
ping: sendto: Network unreachable
/ # cat /etc/resolv.conf
search alkamal.net
nameserver 2001:0000:0000:0000:0000:0000:0000:0001
/ #
```

20 Анализ DHCPv6 Stateful-трафика

- RA с managed-flag
- Solicit → Advertise → Request → Reply
- Адрес успешно выдан

Capturing from Standard input [alkamal-gw-01 eth2 to alkamal-sw-03 Ethernet1]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl>/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000	fe80::42:88ff:fe32:2900	ff02::1:2	ICMPv6	70	Router Solicitation from 02:42:88:32:29:00
2	94.000	fe80::edf:93ff:fe5:2	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
3	94.000	fe80::edf:93ff:fe5:2	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
4	10.000	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	116	Solicit XID: 0xb0281f [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
5	10.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Advertise XID: 0xb0281f IAA: 2001::199 CID: 00030001024288322900
6	10.000	fe80::42:88ff:fe32:2900	ff02::1:ffff::2	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fe5:2 from 02:42:88:32:29:00
7	10.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Advertisement fe80::edf:93ff:fe5:2 (rtr, sol, ovr) is at 0c:df:93:5f:00:02
8	10.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
9	10.000	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	134	Request XID: 0xb0281f [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
10	10.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Reply XID: 0xb0281f IAA: 2001::199 CID: 00030001024288322900
11	10.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
12	10.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Solicitation for fe80::42:88ff:fe32:2900 from 0c:df:93:5f:00:02
13	10.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	78	Neighbor Advertisement fe80::42:88ff:fe32:2900 (sol)
14	13.000	fe80::edf:93ff:fe5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:5f:00:02
15	13.000	fe80::edf:93ff:fe5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:5f:00:02
16	13.000	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	116	Solicit XID: 0x8ad27e [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
17	13.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Advertise XID: 0x8ad27e IAA: 2001::198 CID: 00030001024288322900
18	13.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
19	13.000	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	134	Request XID: 0x8ad27e [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
20	13.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Reply XID: 0x8ad27e IAA: 2001::198 CID: 00030001024288322900
21	13.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
22	13.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fe5:2 from 02:42:88:32:29:00
23	13.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	ICMPv6	78	Neighbor Advertisement fe80::edf:93ff:fe5:2 (rtr, sol)
24	13.000	fe80::edf:93ff:fe5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Solicitation for fe80::42:88ff:fe32:2900 from 0c:df:93:5f:00:02
25	13.000	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fe5:2	ICMPv6	78	Neighbor Advertisement fe80::42:88ff:fe32:2900 (sol)
26	14.000	fe80::edf:93ff:fe5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:5f:00:02
27	14.000	fe80::42:88ff:fe32:2900	2001::1	ICMPv6	118	Echo (ping) request id=0x0157, seq=0, hop limit=64 (reply in 28)
28	14.000	2001::1	fe80::42:88ff:fe32:2900	ICMPv6	118	Echo (ping) reply id=0x0157, seq=0, hop limit=64 (request in 27)
29	14.000	fe80::42:88ff:fe32:2900	2001::1	ICMPv6	118	Echo (ping) request id=0x0157, seq=1, hop limit=64 (reply in 30)
30	14.000	2001::1	fe80::42:88ff:fe32:2900	ICMPv6	118	Echo (ping) reply id=0x0157, seq=1, hop limit=64 (request in 29)